

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 38-2021: A 76-Year-Old Woman with Abdominal Pain, Weight Loss, and Memory Impairment

Lisa L. Willett, M.D., Gabrielle K. Bromberg, M.D., Ryan Chung, M.D.,
 Rebecca K. Leaf, M.D., Rose H. Goldman, M.D., M.P.H., and Amy K. Dickey, M.D.

PRESENTATION OF CASE

From the Department of Medicine, University of Alabama at Birmingham School of Medicine, Birmingham (L.L.W.); and the Departments of Medicine (G.K.B., R.K.L., A.K.D.) and Radiology (R.C.), Massachusetts General Hospital, and the Departments of Medicine (G.K.B., R.K.L., R.H.G., A.K.D.) and Radiology (R.C.), Harvard Medical School, Boston, and the Department of Medicine, Cambridge Health Alliance, Cambridge (R.H.G.) — all in Massachusetts.

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Dr. Gabrielle K. Bromberg: A 76-year-old woman was admitted to this hospital because of abdominal pain, anorexia, weight loss, memory impairment, and suicidal ideation.

The patient had been in her usual state of health until 2 months before this admission, when she began having epigastric pain and anorexia associated with new constipation. Findings on upper and lower endoscopy were reportedly normal, as were the results of serologic testing for celiac disease.

Fifteen days before this admission, the abdominal pain worsened. Ten days before this admission, the patient was admitted to another hospital because of diffuse abdominal pain associated with nausea and poor oral intake. On examination, the patient appeared pale. The abdomen was soft and nontender. The patient was alert and oriented to time, place, and person but had difficulty with word finding. Initial laboratory test results were notable for normocytic anemia and an elevated bilirubin level; other laboratory test results are shown in Table 1.

Dr. Ryan Chung: On admission to the other hospital, computed tomography (CT) of the abdomen and pelvis (Fig. 1A and 1B) was performed after the administration of intravenous contrast material. There was evidence of a previous cholecystectomy, the common bile duct measured 7 mm in diameter (reference range, <10 mm after cholecystectomy), and there was mild anasarca with diffuse mesenteric edema. On magnetic resonance cholangiopancreatography (Fig. 1C and 1D), the common bile duct again measured 7 mm in diameter; findings were otherwise normal.

Dr. Bromberg: After admission to the other hospital, 1 unit of packed red cells was transfused, and the hemoglobin level remained stable thereafter. On the third hospital day, the patient could eat small amounts of food. On further questioning, the patient expressed suicidal ideation and reported a history of prolonged domestic abuse; she was evaluated by the psychiatry service. A diagnosis of post-traumatic stress disorder was made, and lorazepam was administered. The suicidal ideation

resolved, but she was noted to need substantial assistance with self-care. While plans were being made for the patient's discharge to an assisted-living facility, the constipation worsened, and her oral intake decreased. On the seventh hospital day, vomiting developed, and the laboratory evaluation was notable for hyponatremia, with a blood sodium level of 120 mmol per liter (reference range, 135 to 145). A nasogastric tube was placed, and 3% hypertonic saline was administered intravenously. On the eighth day at the other hospital, additional imaging studies were obtained.

Dr. Chung: Repeat CT of the abdomen and pelvis (Fig. 1E and 1F), performed after the administration of intravenous contrast material, revealed multiple dilated, fluid-filled loops of small bowel that measured up to 4 cm in diameter and the presence of air–fluid levels and an abrupt transition point in the right lower quadrant.

Dr. Bromberg: Urine specimens had been sent for measurement of urinary porphyrin levels at the time of admission to the other hospital; the resulting levels were elevated (Table 1), and the patient was transferred to this hospital for further evaluation.

On admission to this hospital, the patient reported persistent abdominal pain, constipation, and anorexia. The vomiting had resolved after insertion of the nasogastric tube. She reported weight loss of 18 kg over the course of the previous 4 months while she was following a limited diet. In an effort to improve the patient's overall health, the patient's daughter had provided her with supplements and recommended a ketogenic diet consisting of protein and cooked vegetables. The patient also reported that she had been having difficulty with memory for 2 months before this admission.

The patient had a history of type 2 diabetes mellitus, hypertension, and glaucoma. Her surgical history included remote cholecystectomy and hysterectomy. Medications included pantoprazole, ondansetron, and calcium carbonate in combination with cholecalciferol. The patient was allergic to ampicillin, penicillin, trimethoprim–sulfamethoxazole, doxycycline, tetracycline, erythromycin, nitrofurantoin, and nortriptyline — all with unknown reactions. She lived with her sister in suburban New England and reported that she was recently separated from her husband

owing to long-standing domestic abuse. The patient had two children. She did not have contact with her son, and her daughter lived in India but visited regularly. Her mother had died from breast cancer. Her father and son both had bipolar affective disorder. The patient did not smoke tobacco, drink alcohol, or use illicit drugs.

On examination, the temperature was 36.1°C, the heart rate 95 beats per minute, the blood pressure 125/55 mm Hg, the respiratory rate 18 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 18.5. The patient appeared cachectic. There was a nasogastric tube in place. The conjunctivae were pale. The abdomen was soft and diffusely tender, without guarding, rigidity, or rebound tenderness. There was no rash. The patient was alert but inconsistently oriented to time, place, and person. Her speech was tangential, and she was unable to recite days of the week backward. The remainder of the examination was normal.

The hemoglobin level was 9.1 g per deciliter (reference range, 12.0 to 16.0). Additional laboratory test results are shown in Table 1. A peripheral-blood smear showed normochromic red cells with occasional burr cells and no schistocytes. Reticulocytes, basophilic stippling, and rare nucleated red cells were present; there were no blasts, and the white cells were morphologically normal.

Dr. Chung: Ultrasonography of the right upper quadrant revealed mildly increased echogenicity and mild coarsening of the liver parenchyma (Fig. 2). The common bile duct measured 5 mm in diameter.

Dr. Bromberg: Intravenous fluids and hemin were administered. On the third hospital day, the volume of output from the nasogastric tube remained elevated. The patient underwent surgical exploration for possible small-bowel obstruction. Dilated loops of small bowel with a transition point near the terminal ileum were observed; there was a loop of small bowel tethered underneath an adhesion band around which the cecum had volvulized. The adhesion was lysed, and the loop of small bowel released. There was no mass or stricture.

A diagnosis was made.

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, Other Hospital	On Admission, Other Hospital	Reference Range, Adults, This Hospital†	On Admission, This Hospital
Blood				
Hemoglobin (g/dl)	12.0–16.0	9.4	14.0–18.0	9.1
Hematocrit (%)	36.0–46.0	28.3	42.0–52.0	27.0
Mean corpuscular volume (fl)	80–100	89	80.0–100.0	91.2
Reticulocyte count (%)	—	—	0.7–2.5	5.7
Platelet count (per μ l)	150,000–450,000	226,000	150,000–450,000	290,000
White-cell count (per μ l)	4500–11,000	6900	4500–11,000	12,030
Differential count (per μ l)				
Neutrophils	—	—	1800–7700	8650
Lymphocytes	—	—	1000–4800	1940
Monocytes	—	—	200–1200	1290
Eosinophils	—	—	0–900	10
Basophils	—	—	0–300	20
Immature granulocytes	—	—	0–100	120
Sodium (mmol/liter)	135–145	133	135–145	131
Potassium (mmol/liter)	3.5–5.5	3.9	3.4–5.0	4.1
Chloride (mmol/liter)	99–113	102	98–108	99
Carbon dioxide (mmol/liter)	20–31	22	23–32	17
Lactic acid (mmol/liter)	—	—	0.5–2.0	1.3
Creatinine (mg/dl)	0.5–0.8	0.9	0.60–1.50	0.97
Urea nitrogen (mg/dl)	9–23	46	8–25	42
Calcium (mg/dl)	8.7–10.4	8.9	8.5–10.5	8.9
Phosphorus (mg/dl)	2.4–5.1	3.0	2.6–4.5	2.6
Magnesium (mg/dl)	1.6–2.6	3.0	1.7–2.4	2.4
Alkaline phosphatase (U/liter)	46–116	65	30–100	81
Alanine aminotransferase (U/liter)	7–40	47	7–33	28
Aspartate aminotransferase (U/liter)	13–40	76	9–32	28
Bilirubin (mg/dl)				
Total	0.3–1.2	2.5	0–1.0	4.8
Direct	0–0.3	0.9	0–0.4	1.8
International normalized ratio	—	1.0	—	—
Albumin (g/dl)	3.2–4.8	4.0	3.3–5.0	4.0
Lactate dehydrogenase (U/liter)	120–246	210	110–210	148
Haptoglobin (mg/dl)	—	—	30–200	129
Ferritin (μ g/liter)	—	—	10–200	1281
Iron (μ g/dl)	—	—	30–160	157
Total iron-binding capacity (μ g/dl)	—	—	230–404	166
Transferrin saturation (%)	—	—	14–50	95
Vitamin B ₁₂ (pg/ml)	—	—	>231	1277
Folate (ng/ml)	—	—	>4.7	15.3

Table 1. (Continued.)

Variable	Reference Range, Adults, Other Hospital	On Admission, Other Hospital	Reference Range, Adults, This Hospital†	On Admission, This Hospital
Glycated hemoglobin (%)	—	—	4.3–6.4	6.2
Urine				
Creatinine, random sample (mg/dl)	—	—	—	100
Osmolality (mOsm/kg water)	—	—	150–1150	787
Sodium (mmol/liter)	—	—	—	<20
Urinary porphyrins and porphyrin precursors, random sample (μg/g creatinine)				
Uroporphyrin I	3.6–21.1	26.8	—	—
Uroporphyrin III	≤5.6	6.6	—	—
Pentacarboxylporphyrins	≤4.1	10.3	—	—
Coproporphyrin I	6.5–33.2	191.9	—	—
Coproporphyrin III	4.8–88.6	3602.0	—	—
Total porphyrins	27.0–153.6	3837.6	—	—

* To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229. To convert the values for magnesium to millimoles per liter, multiply by 0.4114. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for iron and iron-binding capacity to micromoles per liter, multiply by 0.1791. To convert the values for vitamin B₁₂ to picomoles per liter, multiply by 0.7378. To convert the values for folate to nanomoles per liter, multiply by 2.266.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

DIFFERENTIAL DIAGNOSIS

Dr. Lisa L. Willett: This 76-year-old woman had an extensive evaluation with a prolonged hospital stay for chronic progressive gastrointestinal symptoms of generalized abdominal pain, constipation, and weight loss. She also had chronic neurologic symptoms with cognitive decline and memory impairment, as well as suicidal ideation. In addition, she was found to have elevated urinary porphyrin levels. The constellation of her symptoms prompts evaluation for acute porphyria.

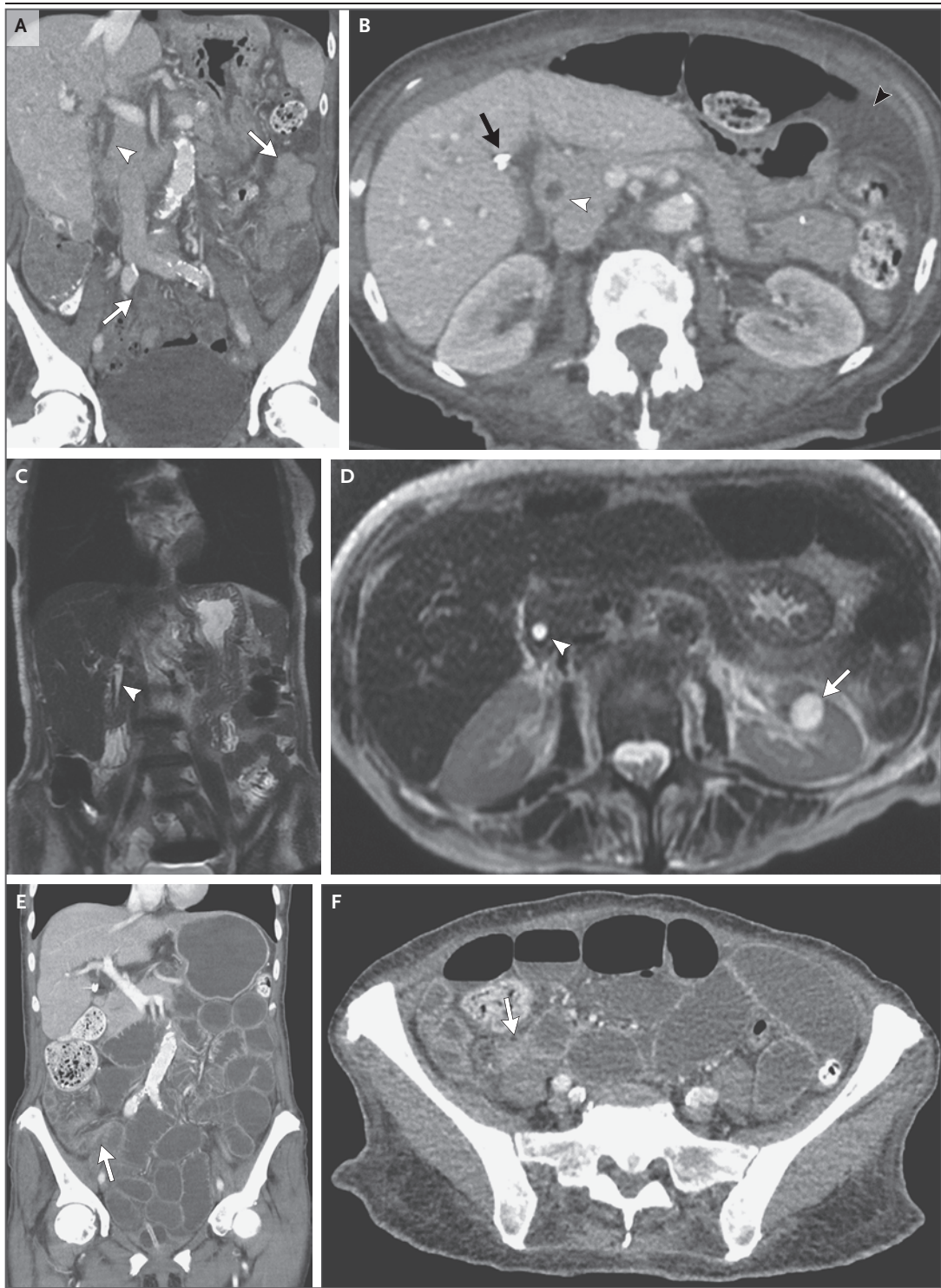
PORPHYRIA

Porphyria refers to a group of metabolic disorders that involve the heme biosynthetic pathway. Eighty percent of heme is produced in the bone marrow, where it combines with globin to form erythroid cells. The remainder of heme is produced in the liver and in cells outside the liver to

make cytochrome P450 enzymes and other proteins.^{1,2}

Heme synthesis begins in the mitochondria with succinyl-coenzyme A and glycine. It involves eight steps, each with a unique enzyme that advances the precursor products along the pathway. The enzyme that initiates the synthesis pathway, delta-aminolevulinic acid synthase (ALAS), differs in the liver (ALAS1) and bone marrow (ALAS2), as does the regulatory feedback control. Disruptions in the pathway resulting from enzyme deficiencies cause precursors to accumulate.^{1,2}

The accumulation of porphyrin precursors (delta-aminolevulinic acid [ALA] or porphobilinogen [PBG]) or porphyrins (i.e., uroporphyrin, coproporphyrin, or protoporphyrin) leads to the clinical features of porphyria.³ Accumulation of the porphyrin precursors causes damage to neurons in the gut, brain, and peripheral nervous system; accumulation of porphyrins causes pho-

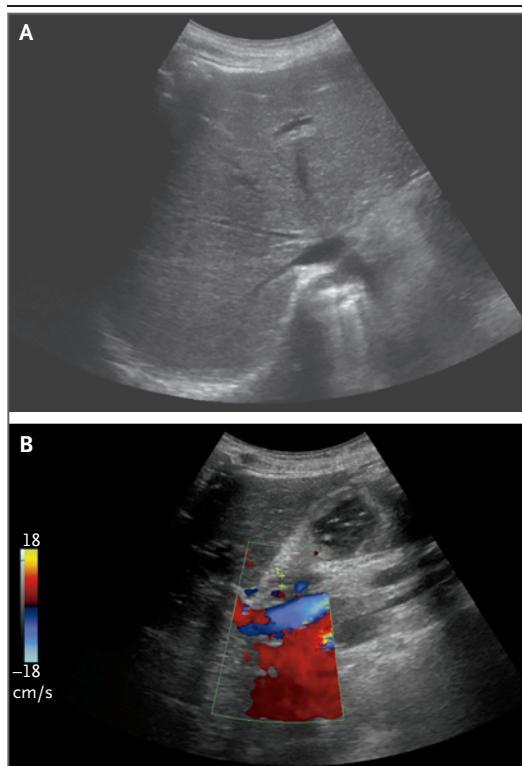


tosensitivity and damage to skin. The specific enzyme deficiency and the corresponding accumulation of precursors lead to the differing patterns of disease in porphyria (Fig. 3).

Porphyria can be categorized as either acute hepatic (neurovisceral) or cutaneous. The four acute hepatic porphyrias are acute intermittent porphyria, variegate porphyria, hereditary cop-

Figure 1 (facing page). Initial Imaging Studies of the Abdomen and Pelvis, Other Hospital.

At the time of admission to the other hospital, coronal (Panel A) and axial (Panel B) images from CT of the abdomen and pelvis, performed after the administration of intravenous contrast material, were obtained. Surgical clips related to a previous cholecystectomy are visible (black arrow), and the common bile duct measures 7 mm in diameter, with smooth distal tapering to the ampulla (white arrowheads). Small-bowel loops of normal caliber are seen throughout the abdomen (white arrows). There is mesenteric edema (black arrowhead) and anasarca (not shown). One day after the initial CT scan, coronal (Panel C) and axial (Panel D) T2-weighted images from magnetic resonance cholangiopancreatography, performed without the administration of intravenous contrast material, were obtained. The common bile duct again measures 7 mm in diameter (arrowheads), without choledocholithiasis. A 1.9-cm simple cyst is seen in the left kidney (arrow). Eight days after the initial CT scan, coronal (Panel E) and axial (Panel F) images from CT of the abdomen and pelvis, performed after the administration of intravenous contrast material, were obtained. There are new dilated, fluid-filled loops of small bowel that are not seen on the initial CT scan; the loops measure up to 4 cm in diameter, with an abrupt transition point in the right lower quadrant (arrows).

**Figure 2. Additional Imaging Studies of the Abdomen, This Hospital.**

Ultrasonography of the right upper quadrant was performed on day 2 at this hospital. A grayscale ultrasound image of the right hepatic lobe (Panel A) shows mildly increased echogenicity and mild coarsening of the liver parenchyma. On a color Doppler image of the hepatic hilum (Panel B, calipers), the common bile duct measures 5 mm in diameter.

porphyria, and ALA dehydratase-deficiency porphyria. Two of these, variegate porphyria and hereditary coproporphyria, have both acute neurovisceral features and cutaneous features.¹ This patient had gastrointestinal and neurologic symptoms; therefore, acute intermittent porphyria must be considered.

ACUTE INTERMITTENT PORPHYRIA

Acute intermittent porphyria is one of the more common types of porphyria.⁴ It is caused by a partial deficiency in the third enzyme of heme synthesis, PBG deaminase. Acute intermittent porphyria is an autosomal dominant disorder, although it has a low penetrance; less than 10% of persons with a pathogenic mutation have symptoms.⁵ The classic presentation of acute intermittent porphyria occurs in young women, with symptoms of fatigue, difficulty concentrating, abdominal pain, nausea, and vomiting. Persons with acute intermittent porphyria can have hypertension, tachycardia, and subtle neurologic signs with an otherwise normal physical examination. Abdominal pain is typically out of proportion to the examination findings, often resulting in a delay in the diagnosis. Common

laboratory findings in patients with acute intermittent porphyria include hyponatremia and mild elevations in liver-enzyme levels. Imaging studies may show an ileus. The diagnosis of acute intermittent porphyria is established on the basis of elevated levels of porphyrins and porphyrin precursors, specifically PBG, in the urine. Elevated levels of ALA and porphyrins are not specific to porphyria; acute hepatic porphyria is diagnosed by substantial elevations in the urinary PBG level. Genetic testing can be useful for confirmation and to establish the type of acute hepatic porphyria. Attacks of acute intermittent porphyria can be precipitated by progesterone surges associated with menses, caloric deprivation, or medications.

This patient had reduced oral intake (initially related to dieting and later to anorexia and nau-

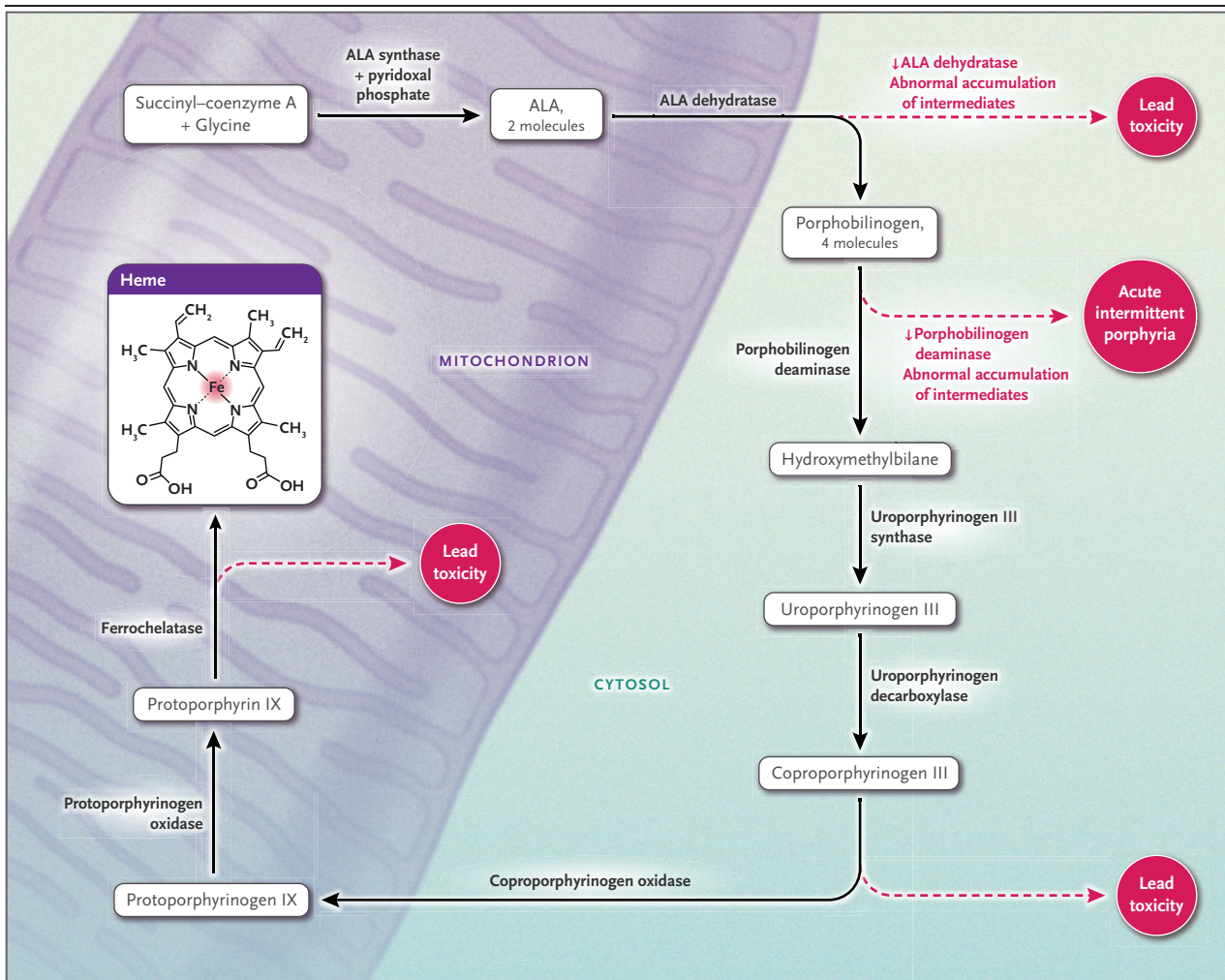


Figure 3. Heme Biosynthetic Pathway.

Shown is the heme biosynthetic pathway, highlighting the enzymes affected in acute intermittent porphyria and lead poisoning. Inhibition of delta-aminolevulinic acid (ALA) dehydratase is thought to be the most important heme pathway enzyme inhibited in lead toxicity.

sea), which could have led to caloric deprivation. Her liver-enzyme levels were initially elevated at the other hospital, and hyponatremia was part of her presentation. Although she had some features of acute intermittent porphyria, her older age, the chronicity of the abdominal pain, and the presence of anemia are not consistent with acute intermittent porphyria and prompt me to seek an alternative diagnosis.

NUTRITIONAL DEFICIENCIES

After learning about the patient's diet, I can expand the differential diagnosis beyond porphyria and consider nutritional deficiencies and side

effects of macronutrients and micronutrients. Macronutrients are major dietary chemical compounds that are needed in large quantities in the body; they include carbohydrates, protein, and fats. Micronutrients refer to vitamins and minerals, which the body needs in small quantities.

This patient had been following a ketogenic diet for months; such a diet could cause both macronutrient and micronutrient deficiency. A ketogenic diet is a high-fat, low-carbohydrate diet that aims to induce ketone production through fat metabolism. The ketogenic diet is being studied as a therapy for many medical illnesses.⁶ Historically, the diet has been used to treat epi-

lepsy, and studies are underway to evaluate its use in Alzheimer's disease, Parkinson's disease, multiple cancers, cardiovascular disease, diabetes, and obesity. A reported side effect of a ketogenic diet is diarrhea, which this patient did not have. The constellation of this patient's signs and symptoms would not be explained by a ketogenic diet, and I am reassured that a nutritional deficiency is not the cause of her illness.

ANEMIA AND ERYTHROPOIESIS

This patient had normocytic anemia, an elevated reticulocyte count, and abnormal findings on a peripheral-blood smear, with nucleated red cells and basophilic stippling. Reticulocytosis is the physiologic response to anemia; the production of nucleated red cells and the presence of basophilic stippling are not.⁷ Nuclear extrusion is the last step before a normoblast enters the circulation to become a reticulocyte. The presence of nucleated red cells on the peripheral-blood smear suggests an abnormality in the bone marrow or in red-cell production. Basophilic stippling is caused by abnormal aggregates of ribosomes and polyribosomes in reticulocytes. Basophilic stippling occurs when hemoglobin synthesis is impaired, usually in the context of thalassemia (impaired globin synthesis) or lead poisoning (impaired heme synthesis). The presence of basophilic stippling is probably a major clue in this case.

LEAD POISONING

Decades ago, lead poisoning was a major public health threat that led to national efforts to protect the public.⁸ Exposure to lead most commonly occurs through occupational exposures associated with the removal of old paint, demolition of buildings, metal mining, or battery manufacturing. In children, accidental ingestion of lead-containing paint causes devastating neurologic consequences. With heightened occupational awareness and preventive measures, lead poisoning is now less common. However, there are many case reports of inadvertent exposure to lead from a variety of sources, such as lead pipes, cookware, dishes, cosmetics, retained bullets, "moonshine" liquor, and herbal supplements. Recently, lead has been found in opium that was sold in the Middle East.⁹ It is also found in Ayurvedic medicines.

Ayurveda is a form of traditional Indian medicine that is used by 80% of India's population and has had increasing popularity in the United States. Rasa shastra is the ancient practice of adding metals, such as lead, arsenic, and mercury, to herbal medicines. Twenty percent of Ayurvedic medicines manufactured in the United States and India contain lead, arsenic, and mercury.¹⁰ Patients may be reluctant to disclose that they are taking these medicines or may not realize that it is relevant medical history. Physicians need to be aware that Ayurvedic medicines, as well as traditional medicines from other cultures, may be a source of heavy-metal poisoning.

Lead poisoning causes a range of nonspecific symptoms, depending on the blood lead level. Adults with blood lead levels between 30 and 50 μg per deciliter can have mild symptoms, including fatigue, irritability, and headache. More severe symptoms are seen with high lead levels (>70 μg per deciliter) and can include anemia, constipation, colicky abdominal pain, and peripheral neuropathy; if the level is higher than 100 μg per deciliter, cerebral edema, encephalopathy, coma, or death can occur. Examination findings are typically nonspecific. Long-term lead exposure may cause Burton's lines on gums, cognitive impairment, and motor weakness in the arms and legs. Common laboratory abnormalities include normocytic or microcytic anemia, basophilic stippling, and abnormalities in kidney or liver function.

Lead poisoning should be considered in any patient presenting with signs and symptoms of porphyria.^{1,2,11} Lead poisoning has been thought to affect several enzymes of heme synthesis, with the primary site of inhibition being ALA dehydratase (Fig. 3).^{2,12} It mimics the clinical presentation of acute intermittent porphyria with abnormal urinary porphyrin levels, as seen in this patient. Lead poisoning could explain the patient's chronic gastrointestinal and neurologic symptoms, the absence of focal examination findings, and the presence of anemia with basophilic stippling. The dilated loops of small bowel were later explained by adhesions resulting from previous surgeries. The patient's daughter visited from India frequently and had provided supplements to the patient in an effort to improve her health. I suspect that this patient had lead poisoning from ingestion of Ayurvedic

medicines and that the diagnostic test was for a blood lead level.

DR. LISA L. WILLETT'S DIAGNOSIS

Lead poisoning from ingestion of Ayurvedic medicines.

DIAGNOSTIC TESTING

Dr. Amy K. Dickey: The urinary porphyrin levels, which were measured at the other hospital, were elevated; the coproporphyrin levels were particularly elevated. Increased coproporphyrin levels can be associated with porphyria but also with other medical conditions, such as heavy-metal poisoning and liver disease, and with certain medications.¹³⁻¹⁷ For many of these causes, the specific porphyrins that are elevated and the degree and mechanism of elevation have not been systematically studied. However, a substantial elevation in the urinary PBG level is specific for the three most common acute hepatic porphyrias.² ALA (an amino acid) and PBG (a pyrrole) are the first two intermediates in the heme biosynthetic pathway, and both accumulate during attacks of these porphyrias.¹⁸ They are not porphyrins, which are tetrapyrroles, and they are not typically included in a urine porphyrin panel.¹⁸ Measurement of the urinary PBG and total porphyrin levels during an acute attack, with the results normalized per gram of urine creatinine, is recommended for screening for acute hepatic porphyrias, and such testing will detect all four acute hepatic porphyrias.¹⁸

This patient was initially treated with intravenous hemin for an acute hepatic porphyria because of reported elevations in the levels of both ALA and PBG. The patient's urinary PBG level was measured at a reference laboratory with the use of a highly sensitive method that provides a low upper limit of the reference range; the level was consequently reported as elevated, at 14.7 μmol per liter (reference value, <1.3). However, after normalizing the value to the creatinine level measured in the same sample and comparing the value to a more typical upper limit, the urinary PBG level was in fact normal, at 3.3 mg per gram of creatinine (reference value, <4.0). In contrast, the urinary ALA level was very elevated, at 83.7 mg per gram of creatinine (reference value, <7.0).¹⁹

During attacks of acute intermittent porphyria, hereditary coproporphyria, and variegate porphyria, the urinary PBG level is expected to be elevated, at 20 to 300 mg per gram of creatinine, accompanied by elevations in the levels of ALA (typically approximately half the PBG level when expressed in mg per gram of creatinine) and total urinary porphyrin.⁴ An elevation in the ALA level but not the PBG level, as was the case in this patient, suggests ALA dehydratase deficiency, which is associated with ALA dehydratase-deficiency porphyria (an extremely rare disease), lead poisoning, and hereditary tyrosinemia.^{20,21} Marked elevations in the levels of urinary coproporphyrin and erythrocyte zinc protoporphyrin are also associated with these conditions.²⁰ ALA dehydratase inhibition can fully explain the elevations in the levels of urinary ALA, coproporphyrin, and erythrocyte zinc protoporphyrin that are seen in persons with lead poisoning.¹⁷

This patient's blood lead level was markedly increased, at 104 μg per deciliter (reference value, <4.9), which confirmed a diagnosis of lead poisoning. The erythrocyte zinc protoporphyrin level was also elevated.²²

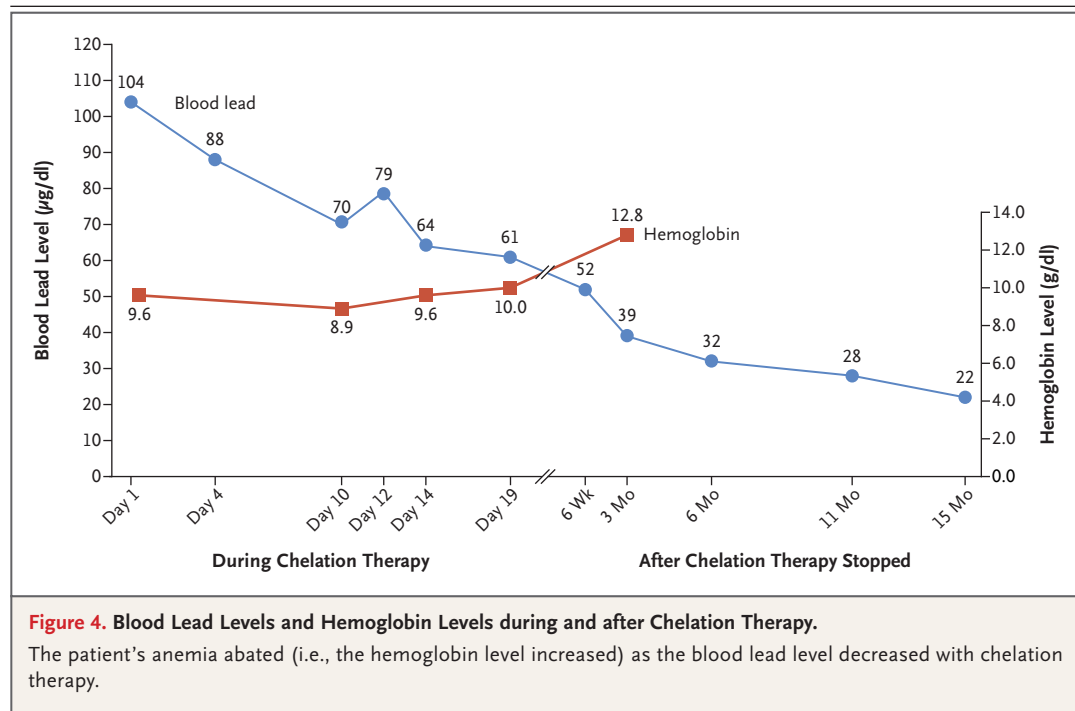
In the diagnosis of rare diseases, two opposite pitfalls are common. The first is not considering the rare disease as a possible diagnosis. For porphyria and lead poisoning, a low threshold for testing is important. The second pitfall is committing to a diagnosis of a rare disease too quickly, as occurred in this case. Misdiagnosis of an acute hepatic porphyria occurs in some patients with abdominal pain and nonspecific laboratory findings, such as reported elevations in urinary ALA and porphyrin levels. This case highlights the importance of performing confirmatory testing, consulting with experts, and being willing to investigate alternative diagnoses.

LABORATORY DIAGNOSIS

Lead poisoning.

DISCUSSION OF MANAGEMENT

Dr. Rebecca K. Leaf: This patient received two primary pharmacologic interventions: hemin when she was initially thought to have an acute hepatic porphyria, and then chelation therapy for lead



poisoning. Chemically, hemin refers to heme (iron protoporphyrin IX) with a coordinating chloride ligand, and in medicine, it is the generic term for heme that is complexed either with a hydroxyl ion (hemin) or with arginine (heme arginate). After intravenous administration, heme is taken up by hepatocytes; it then reconstitutes the regulatory or “free” heme pool that represses synthesis of ALAS1, the rate-limiting enzyme in hepatic heme production.^{23,24} Treatment of an acute hepatic porphyria attack with heme leads to a rapid abatement of symptoms and results in markedly decreased urine PBG excretion.²⁵ Early treatment is most effective and can prevent severe neurologic injury.

Although this patient's symptoms mimicked acute hepatic porphyria and she received 4 days of empirical treatment with heme, the ultimate diagnosis was lead poisoning. The primary strategy for the management of lead poisoning is removal of the source of exposure. However, chelation therapy is recommended for all adult patients with a blood lead level of 100 µg per deciliter or higher, for most adult patients with a blood lead level of 80 µg per deciliter or higher, and for some adult patients with symptoms of lead poisoning and a blood lead level of 50 µg per deciliter or higher.²⁶

Chelation therapy — derived from the Greek word *khele*, or claw — consists of compounds that bind to metal ions and promote their excretion in bile or urine.²⁵ Commonly used lead chelators include calcium disodium EDTA and 2,3-dimercaptosuccinic acid (DMSA). Calcium disodium EDTA is administered parenterally and can cause dose-related nephrotoxicity; DMSA is given orally and can cause transient elevation in aminotransferase levels and skin reactions.²⁷ This patient was treated with DMSA, and follow-up was arranged with an occupational and environmental medicine specialist with experience in treating lead poisoning.

FOLLOW-UP

Dr. Rose H. Goldman: I saw the patient in my clinic on the day of discharge. She appeared frail and displayed some memory issues in relating her history. She had no lead line on her gums, and her abdominal examination was unremarkable.

Six weeks after chelation therapy was stopped, the blood lead level was 52 µg per deciliter (Fig. 4). Since data are lacking to support the notion that faster elimination of lead with the use of chelation therapy would alter any of her long-term risks from lead, and since she had no

current symptoms, she declined further treatment. She reported that her abdominal pain had resolved and that her eating and thinking had improved. Her hemoglobin level was rising.

Fifteen months after chelation therapy was stopped, the blood lead level was 22 μg per deciliter. She was being treated for cognitive impairment in the context of depression, stress, and anxiety related to social and legal issues.

Possible explanations for the patient's persistently elevated blood lead level included continued exposure (which neither she nor her daughter reported), high bone stores in equilibrium with blood (although there had been no known exposures before her use of Ayurvedic medicines), or more likely, mild chronic renal insufficiency with potentially decreased renal excretion of lead. Lead poisoning appeared to be the

major contributor to her anemia; the hemoglobin level increased as the blood lead levels declined (Fig. 4). Lead poisoning can result in permanent effects on cognition, even in adults. The contribution of lead to this patient's cognitive problems is difficult to determine with certainty, given her initial presentation with severe hyponatremia and emotional distress and later, the presence of stress, anxiety, and depression.

FINAL DIAGNOSIS

Lead poisoning.

This case was presented at Medicine Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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